Data stewardship in DH and beyond: promoting responsible, sustainable, and FAIR open research data through education

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Introduction

Relevance

The increasing use of data-driven research in the field of digital humanities has emphasized the fundamental importance of research data management (RDM) and data stewardship skills. Being a digital humanist does not necessarily entail possessing these skills. This abstract highlights the importance of education in these areas to advance open research data, uphold the FAIR principles, and promote sound scientific practices. It also explores how this education can be practically implemented through specialized courses and certifications available at the University of Graz, which demonstrate exemplary practices.

Legal and ethical considerations

Educating scholars about responsible data management includes emphasizing the ethical aspects of collecting, using, and sharing of data, as well as the importance of obtaining proper consent and ensuring privacy protection. Compliance with data protection regulations and intellectual property laws are crucial for safeguarding the integrity of research data and contributing to the trustworthy and responsible advancement of research.

Trustworthiness and community focus

The ability to assess the reliability of research data is essential for researchers, particularly given the rise of artificial intelligence (AI). Evaluating the quality and credibility of underlying data is increasingly important as it significantly impacts subsequent data analysis, visualization, and utilization in AI applications. Researchers need to develop expertise to assess the integrity, accuracy, and representativeness of data, ensuring the soundness of their findings. Additionally, maintaining a community focus is especially crucial in this process, as these skills largely depend on the research domain.

Education Opportunities at the University of Graz

Teaching with practical application

Teaching is led by professionals who possess extensive practical experience. The staff in this field consistently monitor current developments through participation in conferences, advanced training programs, and networking activities. This monitoring occurs at all levels, encompassing internal university initiatives, national engagement (including regular meetings and cooperation with other Austrian institutions and their RDM staff), and international involvement (such as contributions to conferences like FORGE and IDW, participation in DHd and DARIAH-EU working groups, and membership in RDA). Knowledge transfer is realized through two distinct educational opportunities, each offering corresponding credits and qualifications. Additionally, internal staff training sessions are provided, covering specific topics in half-day seminars. For librarians, a specialized elective module named “Data Librarian” is available.

Semester Course “Digital preservation and data management”

This course equips participants with a comprehensive understanding of RDM, metadata and repository management as well as digital long-term preservation. Emphasizing both theoretical concepts and practical exercises, the course addresses the peculiarities of humanities research data (e.g. retro digitization, historical, fuzzy and incomplete datasets, etc.) and offers possible solutions and best practice. The course comprises 4 ECTS and is offered every year. The course is a mandatory component of the DH master’s degree and an elective for all humanities students; the obligatory attendance for DH students fosters the skills needed to provide high quality data.

University Course “Data Stewardship”

Similarly, the "Data Stewardship" certificate course aims to develop participants' skills in handling research data, with a focus on the research data lifecycle, Open Science, FAIR principles, and legal compliance. It comprises 16 ECTS in total and is awarded with a certificate. The course was first held in the summer term 2023 and continues now in its second cycle, constituting the second available certificate in this field in Austria. Even though a large part is dedicated to general concepts, domain specific applications are addressed through lectures with researchers from different disciplines and through student projects that focus on the participant’s own research domains.

Conclusion

Integrating research data management and data stewardship education into academic curricula is essential for advancing open research data, adhering to the FAIR principles, and strengthening good scientific practice. In this way, the digital humanities community can enhance its commitment to openness, transparency, and ethical conduct, ultimately fostering a more sustainable and collaborative research environment. Best results could be achieved by teaching these aspects as early as BA level and intertwining with methodology and research practice in the respective disciplines.

Data management is increasingly essential for not only research institutions but also galleries, libraries, archives and museums, making highly skilled personnel crucial for the digital transformation of research and society. Trustworthy, quality-assured data is becoming increasingly important as the foundation of equally trustworthy and reproducible research results. Ultimately, open research data, supported by public funding, should be complemented by well-educated data stewards and research professionals for maximum benefit to the public and to increase the trust in science and research.

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